Main Ideas of the Level-Set Route

A compact guide to the D_H/D_I proof and the remaining geometric input

Let u be the torsion function of Ω, and write its superlevel sets as

$$E_t = \{u > t\}.$$

We parametrize levels by volume: $t = t(\rho)$ is chosen so that

$$|E_{t(\rho)}| = |B_\rho|, \qquad E_\rho := E_{t(\rho)}.$$

For almost every ρ, E_ρ has finite perimeter and

$$\nu_\rho = -\frac{\nabla u}{|\nabla u|} \qquad \mathcal{H}^{n-1}\text{-a.e. on } \partial^* E_\rho.$$

Set

$$w(\rho) := -t'(\rho), \qquad d\mu(\rho) = w(\rho)\, d\rho, \qquad V_\rho := \frac{w(\rho)}{|\nabla u|}.$$

Thus V_ρ is the normal velocity of the level set under the ρ-flow. The corresponding average ball velocity is

$$\bar{V}_\rho := \frac{P(B_\rho)}{P(E_\rho)}.$$

The coarea identities give

$$\int_{\partial^* E_\rho} V_\rho \, d\mathcal{H}^{n-1} = P(B_\rho), \qquad \bar{V}_\rho = \frac{1}{P(E_\rho)} \int_{\partial^* E_\rho} V_\rho \, d\mathcal{H}^{n-1}.$$

The two defects

The new proof gives an exact budget in terms of two nonnegative level defects. The first is the Cauchy–Schwarz defect

$$D_H(t(\rho)) := \left(\int_{\partial^* E_\rho} \frac{1}{|\nabla u|} \, d\mathcal{H}^{n-1} \right) \left(\int_{\partial^* E_\rho} |\nabla u| \, d\mathcal{H}^{n-1} \right) - P(E_\rho)^2.$$

It vanishes exactly when $|\nabla u|$ is constant on the level. The second is the squared-perimeter isoperimetric defect

$$D_I(t(\rho)) := P(E_\rho)^2 - P(B_\rho)^2.$$

It vanishes exactly when E_ρ is a ball.

The exact level-set identity gives, on good intervals $J \subset [\rho_*, \rho_\delta]$,

$$\int_J \big(D_H(t(\rho)) + D_I(t(\rho)) \big) \, d\mu(\rho) \le C(n, R, \rho_*) \delta_T(\Omega).$$

Since both defects are nonnegative, each has the same average bound separately.

The interpretation is simple. The defect D_I controls how round the levels are. Quantitative isoperimetry gives

$$q(\rho)^2 \le C(n, \rho_*) D_I(t(\rho)), \qquad q(\rho) := \inf_{z \in \mathbb{R}^n} |E_\rho \Delta B_\rho(z)|.$$

The defect D_H controls how uniformly the level moves, because

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| \, d\mathcal{H}^{n-1} \right)^2 \le D_H(t(\rho)).$$

Thus D_I pays for shape, while D_H pays for speed.

The moving-ball distance

The quantity propagated to the endpoint is

$$q(\rho) := \inf_{z\in\mathbb{R}^n} |E_\rho \Delta B_\rho(z)|.$$

It is Lipschitz for a reason that uses only nesting:

$$|q(\sigma) - q(\rho)| \le 2\omega_n(\sigma^n - \rho^n) \le 2n\omega_n|\sigma - \rho|.$$

The average bound $\int q^2\, d\mu \le C\delta_T(\Omega)$ follows from D_I, but this alone does not control the endpoint. A function can be small on average and still rise near the end. The first variation of q prevents this rise unless the deficit pays for it.

Fix a good ρ, and let z_ρ minimize $z \mapsto |E_\rho \Delta B_\rho(z)|$. Compare $E_{\rho+h}$ with an affine image of E_ρ which sends the comparison ball exactly to $B_{\rho+h}(z_\rho + ha)$. On the boundary, the mismatch between the true level motion and this affine motion is

$$V_\rho - H_{z_\rho,\rho} - a\cdot\nu_\rho, \qquad H_{z,\rho}(x) := \frac{(x-z)\cdot\nu_\rho(x)}{\rho}.$$

This gives the one-sided variation formula

$$D^+q(\rho) \le \frac{n}{\rho}q(\rho) + \inf_{a\in\mathbb{R}^n}\int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho} - a\cdot\nu_\rho|\, d\mathcal{H}^{n-1}.$$

For the current route, take $a = 0$ and insert $\bar{V}_\rho$:

$$D^+q(\rho) \le \frac{n}{\rho}q(\rho) + \int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho|\, d\mathcal{H}^{n-1} + \int_{\partial^* E_\rho} |H_{z_\rho,\rho} - \bar{V}_\rho|\, d\mathcal{H}^{n-1}.$$

The first term is controlled by D_I, the second by D_H, and the third is the remaining geometric trace term.

If the trace term satisfies

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho,\rho} - \bar{V}_\rho|\, d\mathcal{H}^{n-1}\right)^2 \le C(n,R,\rho_*)D_I(t(\rho)),$$

then squaring and integrating gives the positive kinetic estimate

$$\int_J (q'_+(\rho))^2\, d\rho \le C(n,R,\rho_*)\delta_T(\Omega), \qquad q'_+ = \max\{q', 0\}.$$

Bad radii are harmless because q is unconditionally Lipschitz and the bad set has measure $O(\delta_T(\Omega))$.

Finally, the average bound for q^2 gives a radius ρ_0 near the endpoint with $q(\rho_0)^2 \le C\delta_T(\Omega)$. Then

$$q(\rho_\delta) \le q(\rho_0) + \int_{\rho_0}^{\rho_\delta} q'_+(\rho)\, d\rho,$$

so Cauchy–Schwarz and the kinetic estimate imply

$$q(\rho_\delta)^2 \le C\delta_T(\Omega).$$

This is the endpoint asymmetry estimate.

The geometric input

The missing input is a same-centre estimate. For good levels, one needs a centre z_ρ such that

$$|E_\rho \Delta B_\rho(z_\rho)|^2 + \int_{\partial^* E_\rho} |\nu_\rho - e_{z_\rho}|^2\, d\mathcal{H}^{n-1} \le C(n,R,\rho_*)D_I(t(\rho)), \qquad e_z(x) = \frac{x-z}{|x-z|}.$$

The radial trace lemma then turns this into the needed estimate for $H_{z_\rho,\rho} - \bar{V}_\rho$.

There is a centre issue. The variation formula uses a centre minimizing $|E_\rho \Delta B_\rho(z)|$. The FMP strategy below naturally selects a potential-maximizing centre. The proof must either show the trace estimate directly for a minimizing centre, or prove it for the FMP centre and then transfer it to a minimizing centre. The manuscript already has the right transfer mechanism: if two same-radius balls both approximate E_ρ, their centres are close except on radii whose measure is paid for by Chebyshev.

The FMP/capacitary strategy

For a finite-perimeter set E with $|E| = |B_\rho|$, define the FMP oscillation index

$$\beta(E)^2 := P(E) - (n-1)\sup_y \int_E |x-y|^{-1}\,dx.$$

Fusco–Maggi–Pratelli gives

$$\beta(E)^2 \le C(n)(P(E) - P(B_\rho)).$$

Choose z maximizing the potential and set

$$\mathcal{C}_z(E) := (n-1)\int_{B_\rho(z)} |x-z|^{-1}\,dx - (n-1)\int_E |x-z|^{-1}\,dx.$$

Because $(n-1)\int_{B_\rho(z)} |x-z|^{-1}\,dx = P(B_\rho)$,

$$\beta(E)^2 = P(E) - P(B_\rho) + \mathcal{C}_z(E).$$

Thus $\mathcal{C}_z(E)$ is controlled by the perimeter deficit. A BV Gauss–Green identity gives

$$\frac{1}{2}\int_{\partial^* E} |\nu_E - e_z|^2\,d\mathcal{H}^{n-1} = P(E) - P(B_\rho) + \mathcal{C}_z(E),$$

so the radial normal error is also controlled.

It remains to prove the elementary shell lemma

$$|E\Delta B_\rho(z)|^2 \le C(n,R,\rho_*)\mathcal{C}_z(E).$$

Write $A = B_\rho(z)\setminus E$ and $F = E\setminus B_\rho(z)$. Since $|A| = |F|$, the potential deficit is the loss from moving mass from inside the ball to outside the ball. By the bathtub principle, the cheapest such move removes mass from the shell just inside $\partial B_\rho(z)$ and adds it to the shell just outside. The first-order terms cancel, and the second-order shell calculation gives a quadratic loss in the exchanged volume. This gives the desired same-centre volume estimate.

Combining FMP, radial Gauss–Green, and the shell lemma yields the same-centre package from D_I alone. That closes the geometric trace term; together with D_H controlling velocity, the variation argument for q closes the quantitative Faber–Krahn route.